

Zinan Lin

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Employment

Microsoft Research

Senior Researcher

Redmond, WA, USA

Oct. 2022-Present

Education

Carnegie Mellon University

Ph.D., Department of Electrical and Computer Engineering

Secondary Master in Machine Learning, School of Computer Science

Secondary Master in Electrical and Computer Engineering

Ph.D. advisors: Giulia Fanti and Vyas Sekar

Grade: 4.0/4.0

Pittsburgh, PA, USA

2017–2023

Tsinghua University

Bachelor of Engineering, Department of Electronic Engineering

Grade: 92/100. Rank: 5/195

Beijing, China

2013–2017

Honors and Awards

12th Annual Best Scientific Cybersecurity Paper, with Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Yu Cheng, Sanmi Koyejo, Dawn Song, Bo Li, granted by the National Security Agency (NSA), <https://www.nsa.gov/Press-Room/News-Highlights/Article/Article/3990748/nsa-awards-authors-of-assessment-of-trustworthiness-in-gpt-models/>

2024

Best Machine Learning and Security Paper (Cybersecurity Best Paper Award), with Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Yu Cheng, Sanmi Koyejo, Dawn Song, Bo Li, granted by SpringerOpen, <https://cybersecurity.springeropen.com/award-2024>

2024

ICML Spotlight (Top 3.5%), with Chulin Xie, Arturs Backurs, Sivakanth Gopi, Da Yu, Huseyin Inan, Harsha Nori, Haotian Jiang, Huishuai Zhang, Yin Tat Lee, Bo Li, Sergey Yekhanin

2024

NeurIPS Outstanding Paper & Oral (Top 0.045%), with Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Yu Cheng, Sanmi Koyejo, Dawn Song, Bo Li

2023

NeurIPS Oral (Synthetic Data Generation with Generative AI Workshop), with Sivakanth Gopi, Janardhan Kulkarni, Harsha Nori, Sergey Yekhanin 2023

Top Reviewers (Top 2.6%) in NeurIPS 2023, <https://nips.cc/Conferences/2023/ProgramCommittee> 2023

AAAI Scholarship, granted by Association for the Advancement of Artificial Intelligence 2022

Outstanding Reviewer (Top 8%) in NeurIPS 2021, <https://nips.cc/Conferences/2021/ProgramCommittee> 2021

IMC Best Paper Finalist, with Alankar Jain, Chen Wang, Giulia Fanti, Vyas Sekar 2020

Top Reviewers (Top 7.6%) in ICML 2020, <https://icml.cc/Conferences/2020/Reviewers> 2020

Cylab Presidential Fellowship, granted by Carnegie Mellon University 2020

Siemens FutureMakers Fellowship, granted by Siemens 2019

Best Reviewers (Top 400) in NeurIPS 2019, <https://nips.cc/Conferences/2019/Reviewers> 2019

NeurIPS Spotlight (Top 4.1%), with Kiran Thekumparampil, Ashish Kheta, and Sewoong Oh 2018

Presidential Fellowship, granted by Carnegie Mellon University 2017

Carnegie Institute of Technology Dean's Fellow, granted by Carnegie Mellon University 2017

Outstanding Bachelor Thesis, granted by Tsinghua University 2017

Meritorious Winner (9% Worldwide), COMAP's Math Contest in Modeling 2015, 2016, 2017

National Scholarship, granted by the government of China 2014, 2015, 2016

Tsinghua Spark Class Fellowship (Top 1%), for top students on scientific research 2015

The First Prize, National Physics Contest for College Student 2014

Experience

NVIDIA (Research Intern) <i>Host: Ming-Yu Liu, Xun Huang</i> ○ Topic: Denoising Diffusion Probabilistic Models (DDPM)	Santa Clara, CA, USA May 2021–Dec. 2021
Google (Research Intern) <i>Host: Yundi Qian</i> ○ Topic: compiler optimizations with reinforcement learning	Mountain View, CA, USA May 2020–Aug. 2020
Microsoft Research Asia (Research Intern) <i>Managers: Fei Gao, Taifeng Wang</i> ○ Topic: a large-scale empirical study of optimization methods	Beijing, China Mar. 2017–Jun. 2017
Luogu Website (Cofounder and Developer) https://www.luogu.com.cn/ ○ One of the biggest online judges in China.	China 2013

Skills

Programming Languages.....
C, C++, Python, Java, (Visual) Basic, Pascal, Haskell, MATLAB, Mathematica, PHP, JavaScript, HTML, CSS, SQL, Verilog, Assembly, bash, shell, L^AT_EX, etc.

Machine Learning Frameworks.....

TensorFlow, PyTorch, Theano, Keras, Blocks, CNTK, etc.

Teaching Assistant

CMU 18752: Estimation, Detection and Learning

Instructor: Rohit Negi

Pittsburgh, PA, USA

Spring 2020, Spring 2021

Services

I am serving as a reviewer/PC member for:

Conference on Neural Information Processing Systems (NeurIPS) 2019, 2020, 2021, 2022, 2023

International Conference on Machine Learning (ICML) 2020, 2021, 2022, 2023, 2024

International Conference on Learning Representations (ICLR) 2021, 2022, 2023, 2024, 2025

Computer Vision and Pattern Recognition Conference (CVPR) 2022, 2023, 2024

International Conference on Computer Vision (ICCV) 2023

European Conference on Computer Vision (ECCV) 2022, 2024

International Symposium on Information Theory (ISIT) 2020, 2021

Artificial Intelligence and Statistics (AISTATS) 2021

ICAIF Workshop on Synthetic Data for AI in Finance 2022

ACM CoNEXT NativeNI Workshop 2022

NeurIPS SyntheticData4ML Workshop 2022

Transactions on Machine Learning Research

Transactions on Pattern Analysis and Machine Intelligence

IET Image Processing

Transactions on Dependable and Secure Computing

Neural Networks

IEEE Access

IEEE/ACM Transactions on Networking

IEEE Transactions on Big Data

International Journal of Intelligent Systems

Publications (* denotes equal contribution, + denotes co-advising)

- [1] Xuefei Ning*, Zifu Wang*, Shiyao Li*, **Zinan Lin***, Peiran Yao*, Tianyu Fu, Matthew B Blaschko, Guohao Dai, Huazhong Yang, and Yu Wang. "Can LLMs Learn by Teaching for Better Reasoning? A Preliminary Study". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2024). URL: <https://arxiv.org/abs/2406.14629>.

- [2] Sangyun Lee, **Zinan Lin**, and Giulia Fanti. “Improving the Training of Rectified Flows”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2024). URL: <https://arxiv.org/abs/2405.20320>.
- [3] Chengquan Guo*, Xun Liu*, Chulin Xie*, Andy Zhou, Yi Zeng, **Zinan Lin**, Dawn Song, and Bo Li. “RedCode: Risky Code Execution and Generation Benchmark for Code Agents”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2024). URL: <https://arxiv.org/abs/2411.07781>.
- [4] Kaiyi Huang, Yukun Huang, Xuefei Ning, **Zinan Lin**, Yu Wang, and Xihui Liu. “GenMAC: Compositional Text-to-Video Generation with Multi-Agent Collaboration”. In: *arXiv preprint arXiv:2412.04440* (2024). URL: <https://arxiv.org/abs/2412.04440>.
- [5] Enshu Liu*, Junyi Zhu*, **Zinan Lin**+, Xuefei Ning+, Matthew B Blaschko, Sergey Yekhanin, Shengen Yan, Guohao Dai, Huazhong Yang, and Yu Wang. “Linear Combination of Saved Checkpoints Makes Consistency and Diffusion Models Better”. In: *arXiv preprint arXiv:2404.02241* (2024). URL: <https://arxiv.org/abs/2404.02241>.
- [6] Enshu Liu*, Junyi Zhu*, **Zinan Lin**+, Xuefei Ning+, Matthew B Blaschko, Shengen Yan, Guohao Dai, Huazhong Yang, and Yu Wang. “Efficient Expert Pruning for Sparse Mixture-of-Experts Language Models: Enhancing Performance and Reducing Inference Costs”. In: *arXiv preprint arXiv:2407.00945* (2024). URL: <https://arxiv.org/abs/2407.00945>.
- [7] **Zinan Lin**, Sivakanth Gopi, Janardhan Kulkarni, Harsha Nori, and Sergey Yekhanin. “Differentially Private Synthetic Data via Foundation Model APIs 1: Images”. In: *International Conference on Learning Representations (ICLR)* (2024). URL: <https://arxiv.org/abs/2305.15560>.
- [8] Chulin Xie, **Zinan Lin**, Arturs Backurs, Sivakanth Gopi, Da Yu, Huseyin Inan, Harsha Nori, Haotian Jiang, Huishuai Zhang, Yin Tat Lee, Bo Li, and Sergey Yekhanin. “Differentially Private Synthetic Data via Foundation Model APIs 2: Text”. In: *International Conference on Machine Learning (ICML)* (2024). URL: <https://arxiv.org/abs/2403.01749>.
- [9] Xuefei Ning*, **Zinan Lin***, Zixuan Zhou*, Huazhong Yang, and Yu Wang. “Skeleton-of-Thought: Prompting LLMs for Efficient Parallel Generation”. In: *International Conference on Learning Representations (ICLR)* (2024). URL: <https://arxiv.org/abs/2307.15337>.
- [10] Shuaiqi Wang, **Zinan Lin**, and Giulia Fanti. “Statistic Maximal Leakage”. In: *The IEEE International Symposium on Information Theory (ISIT)* (2024). URL: <https://arxiv.org/abs/2411.18531>.
- [11] Xinyu Tang, Richard Shin, Huseyin A. Inan, Andre Manoel, Fatemehsadat Mireshghallah, **Zinan Lin**, Sivakanth Gopi, Janardhan Kulkarni, and Robert Sim. “Privacy-Preserving In-Context Learning with Differentially Private Few-Shot Generation”. In: *International Conference on Learning Representations (ICLR)* (2024). URL: <https://arxiv.org/abs/2309.11765>.
- [12] Arturs Backurs, **Zinan Lin**, Sepideh Mahabadi, Sandeep Silwal, and Jakub Tarnawski. “Efficiently Computing Similarities to Private Datasets”. In: *International Conference on Learning Representations (ICLR)* (2024). URL: <https://arxiv.org/abs/2403.08917>.
- [13] Junyi Zhu, **Zinan Lin**, Enshu Liu, Xuefei Ning, and Matthew B. Blaschko. “Rescaling Intermediate Features Makes Trained Consistency Models Perform Better”. In: *International Conference on Learning Representations (ICLR) (TinyPapers)* (2024). URL: <https://openreview.net/forum?id=1o3LnwflAl>.

- [14] Ronghao Ni, **Zinan Lin**, Shuaiqi Wang, and Giulia Fanti. “Mixture-of-Linear-Experts for Long-term Time Series Forecasting”. In: *International Conference on Artificial Intelligence and Statistics (AISTATS)* (2024). URL: <https://arxiv.org/abs/2312.06786>.
- [15] Lin Zhao*, Tianchen Zhao*, **Zinan Lin**, Xuefei Ning, Guohao Dai, Huazhong Yang, and Yu Wang. “FlashEval: Towards Fast and Accurate Evaluation of Text-to-image Diffusion Generative Models”. In: *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2024). URL: <https://arxiv.org/abs/2403.16379>.
- [16] **Zinan Lin***, Shuaiqi Wang*, Vyas Sekar, and Giulia Fanti. “Summary Statistic Privacy in Data Sharing”. In: *IEEE Journal on Selected Areas in Information Theory (JSAIT)* (2024). URL: <https://arxiv.org/abs/2303.02014>.
- [17] Da Yu, Sivakanth Gopi, Janardhan Kulkarni, **Zinan Lin**, Saurabh Naik, Tomasz Lukasz Religa, Jian Yin, and Huishuai Zhang. “Selective Pre-training for Private Fine-tuning”. In: *Transactions on Machine Learning Research (TMLR)* (2024). URL: <https://arxiv.org/abs/2305.13865>.
- [18] Tianchen Zhao, Xuefei Ning, Tongcheng Fang, Enshu Liu, Guyue Huang, **Zinan Lin**, Shengen Yan, Guohao Dai, and Yu Wang. “MixDQ: Memory-Efficient Few-Step Text-to-Image Diffusion Models with Metric-Decoupled Mixed Precision Quantization”. In: *European Conference on Computer Vision (ECCV)* (2024). URL: <https://arxiv.org/abs/2405.17873>.
- [19] Tianchen Zhao, Tongcheng Fang, Enshu Liu, Wan Rui, Widyadewi Soedarmadji, Shiyao Li, **Zinan Lin**, Guohao Dai, Shengen Yan, Huazhong Yang, et al. “ViDiT-Q: Efficient and Accurate Quantization of Diffusion Transformers for Image and Video Generation”. In: *arXiv preprint arXiv:2406.02540* (2024). URL: <https://arxiv.org/abs/2406.02540>.
- [20] Shuaiqi Wang, Shuran Zheng, **Zinan Lin**, Giulia Fanti, and Zhiwei Steven Wu. “Inferentially-Private Private Information”. In: *arXiv preprint arXiv:2410.17095* (2024). URL: <https://arxiv.org/abs/2406.02540>.
- [21] Enshu Liu*, Xuefei Ning*, **Zinan Lin***, Huazhong Yang, and Yu Wang. “OMS-DPM: Optimizing the Model Schedule for Diffusion Probabilistic Models”. In: *International Conference on Machine Learning (ICML)*. 2023. URL: <https://arxiv.org/abs/2306.08860>.
- [22] Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, **Zinan Lin**, Yu Cheng, Sanmi Koyejo, Dawn Song, and Bo Li. “DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. URL: <https://arxiv.org/abs/2306.11698>.
- [23] Da Yu, Arturs Backurs, Sivakanth Gopi, Huseyin Inan, Janardhan Kulkarni, **Zinan Lin**, Chulin Xie, Huishuai Zhang, and Wanrong Zhang. “Training Private and Efficient Language Models with Synthetic Data from LLMs”. In: *NeurIPS Workshop on Socially Responsible Language Modelling Research* (2023). URL: <https://openreview.net/forum?id=FKwtKzglFb>.
- [24] **Zinan Lin**. “Data Sharing with Generative Adversarial Networks: From Theory to Practice”. PhD thesis. Carnegie Mellon University, 2022. URL: https://zinanlin.me/bio/PhD_thesis.pdf.
- [25] **Zinan Lin***, Shuaiqi Wang*, Vyas Sekar, and Giulia Fanti. “Distributional Privacy for Data Sharing”. In: *NeurIPS 2022 Workshop on Synthetic Data for Empowering ML Research*. URL: <https://openreview.net/forum?id=6oVAzFsHLFK>.

- [26] Yucheng Yin, **Zinan Lin**, Minhao Jin, Giulia Fanti, and Vyas Sekar. “Practical GAN-based Synthetic IP Header Trace Generation using NetShare”. In: *ACM Special Interest Group on Data Communication (SIGCOMM)*. 2022. URL: <https://dl.acm.org/doi/10.1145/3544216.3544251>.
- [27] **Zinan Lin**, Hao Liang, Giulia Fanti, and Vyas Sekar. “RareGAN: Generating Samples for Rare Classes”. In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2022. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/20715>.
- [28] Yucheng Yin, **Zinan Lin**, Minhao Jin, Giulia Fanti, and Vyas Sekar. “PcapShare: Exploring the Feasibility of GANs for Synthetic Packet Header Trace Generation”. In: *International Conference on COMMunication Systems and NETWORKS (COMSNETS) (demo)*. 2022. URL: https://www.comsnets.org/demos_exhibits.html.
- [29] **Zinan Lin**, Vyas Sekar, and Giulia Fanti. “Why Spectral Normalization Stabilizes GANs: Analysis and Improvements”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2021. URL: <http://arxiv.org/abs/2009.02773>.
- [30] **Zinan Lin**, Vyas Sekar, and Giulia Fanti. “On the Privacy Properties of GAN-generated Samples”. In: *International Conference on Artificial Intelligence and Statistics (AISTATS)*. PMLR. 2021, pp. 1522–1530. URL: <https://arxiv.org/abs/2206.01349>.
- [31] Todd Huster, Jeremy E.J. Cohen, **Zinan Lin**, Kevin Chan, Cho-Yu Jason Chiang, and Vyas Sekar. “Pareto GAN: Extending the Representational Power of GANs to Heavy-Tailed Distributions”. In: *International Conference on Machine Learning (ICML)*. 2021. URL: <http://proceedings.mlr.press/v139/huster21a.html>.
- [32] Mircea Trofin, Yundi Qian, Eugene Brevdo, **Zinan Lin**, Krzysztof Choromanski, and David Li. “MLGO: a Machine Learning Guided Compiler Optimizations Framework”. In: *arXiv preprint arXiv:2101.04808* (2021). URL: <https://arxiv.org/abs/2101.04808>.
- [33] **Zinan Lin**, Kiran Koshy Thekumparampil, Giulia Fanti, and Sewoong Oh. “InfoGAN-CR and ModelCentrality: Self-supervised Model Training and Selection for Disentangling GANs”. In: *International Conference on Machine Learning (ICML)*. 2020, pp. 7775–7786. URL: <https://arxiv.org/abs/1906.06034>.
- [34] **Zinan Lin**, Alankar Jain, Chen Wang, Giulia Fanti, and Vyas Sekar. “Using GANs for Sharing Networked Timeseries Data: Challenges, Initial Promise, and Open Questions”. In: *Proceedings of the Internet Measurement Conference (IMC)*. 2020. URL: <http://arxiv.org/abs/1909.13403>.
- [35] **Zinan Lin**, Ashish Khetan, Giulia Fanti, and Sewoong Oh. “PacGAN: The Power of Two Samples in Generative Adversarial Networks”. In: *IEEE Journal on Selected Areas in Information Theory (JSAIT)* 1.1 (2020), pp. 324–335. URL: <https://ieeexplore.ieee.org/document/9046238>.
- [36] **Zinan Lin**, Soo-Jin Moon, Carolina M. Zarate, Ritika Mulagalapalli, Sekar Kulandaivel, Giulia Fanti, and Vyas Sekar. “Towards Oblivious Network Analysis using Generative Adversarial Networks”. In: *Proceedings of the 18th ACM Workshop on Hot Topics in Networks (HotNets)*. ACM. 2019. URL: <https://dl.acm.org/doi/10.1145/3365609.3365854>.
- [37] **Zinan Lin**, Ashish Khetan, Giulia Fanti, and Sewoong Oh. “PacGAN: The Power of Two Samples in Generative Adversarial Networks”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2018, pp. 1498–1507. URL: <https://arxiv.org/abs/1712.04086>.

- [38] Kiran K Thekumparampil, Ashish Khetan, **Zinan Lin**, and Sewoong Oh. “Robustness of Conditional GANs to Noisy Labels”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2018, pp. 10271–10282. URL: <https://arxiv.org/abs/1811.03205>.
- [39] **Zinan Lin**, Yongfeng Huang, and Jilong Wang. “RNN-SM: Fast Steganalysis of VoIP Streams Using Recurrent Neural Network”. In: *IEEE Transactions on Information Forensics and Security (TIFS)* 13.7 (July 2018), pp. 1854–1868. ISSN: 1556-6013. DOI: 10.1109/TIFS.2018.2806741. URL: <http://ieeexplore.ieee.org/document/8292900>.